AREA — Short Revision Notes

Apply Level: Use Knowledge in New Situations

1 Finding Area of Square and Rectangle

Shape	Formula	Example
Square	$A = a^2$	Side = 5 cm → A = 25 cm^2
Rectangle	$A = l \times b$	l = 10 cm, b = 6 cm → A = 60 cm^2

Step-by-Step Examples

Problem	Step 1	Step 2	Answer
Square side 7 cm	a = 7	$A = 7^2$	**49 cm^2**
Rectangle length 12 cm, breadth 5 cm	l = 12, b = 5	A = 12×5	**60 cm^2**
Square side 8 m	a = 8	$A = 8^2$	**64 m^2**
Rectangle length 15 cm, breadth 8 cm	l = 15, b = 8	A = 15×8	**120 cm^2**

Quick Application

Shape	Measurements	Area
Square	side = 6 cm	**36 cm^2**
Square	side = 9 m	**81 m^2**
Rectangle	l = 8 cm, b = 4 cm	**32 cm^2**
Rectangle	l = 20 cm, b = 10 cm	**200 cm^2**
Rectangle	l = 3.5 m, b = 2 m	**7 m^2**

2 Finding Area of Triangle

Formula: A = ½ × base × height

Problem	Step 1	Step 2	Answer
base=12 cm, height=5 cm	½ × 12 × 5	½ × 60	**30 cm^2**
base=10 cm, height=8 cm	½ × 10 × 8	½ × 80	**40 cm^2**
base=14 m, height=6 m	½ × 14 × 6	½ × 84	**42 m^2**
base=8 cm, height=7 cm	½ × 8 × 7	½ × 56	**28 cm^2**

Quick Application

Base	Height	Area
10 cm	6 cm	**30 cm²**
15 cm	4 cm	**30 cm²**
20 cm	5 cm	**50 cm²**
12 m	9 m	**54 m²**
16 cm	8 cm	**64 cm²**

Finding Height Given Area

Problem	Step 1	Step 2	Answer
Area=48 cm², base=12 cm	48 = ½ × 12 × h	48 = 6h	**h = 8 cm**
Area=60 cm², base=15 cm	60 = ½ × 15 × h	60 = 7.5h	**h = 8 cm**

3 Finding Area of Parallelogram

`Formula: A = base × height`

Problem	Step 1	Step 2	Answer
base=10 cm, height=5 cm	10 × 5	—	**50 cm²**
base=8 m, height=4 m	8 × 4	—	**32 m²**
base=12 cm, height=7 cm	12 × 7	—	**84 cm²**
base=15 m, height=10 m	15 × 10	—	**150 m²**

Quick Application

Base	Height	Area
7 cm	4 cm	**28 cm²**
9 cm	6 cm	**54 cm²**
11 cm	5 cm	**55 cm²**
14 m	8 m	**112 m²**
20 cm	3 cm	**60 cm²**

Finding Height Given Area

Problem	Step 1	Step 2	Answer
Area=48 cm², base=8 cm	48 = 8 × h	h = 48÷8	**h = 6 cm**
Area=105 cm², base=7 cm	105 = 7 × h	h = 105÷7	**h = 15 cm**

4 Finding Area of Trapezium

`Formula: A = ½ × (sum of parallel sides) × height`

Problem	Step 1	Step 2	Step 3	Answer
a=11 cm, b=6 cm, h=8 cm	sum = 11+6=17	½ × 17 × 8	½ × 136	**68 cm^2**
a=12 cm, b=8 cm, h=7 cm	sum = 12+8=20	½ × 20 × 7	½ × 140	**70 cm^2**
a=15 m, b=10 m, h=6 m	sum = 15+10=25	½ × 25 × 6	½ × 150	**75 m^2**
a=9 cm, b=5 cm, h=4 cm	sum = 9+5=14	½ × 14 × 4	½ × 56	**28 cm^2**

Quick Application

Side a	Side b	Height	Area
8 cm	6 cm	5 cm	**35 cm^2**
10 cm	4 cm	6 cm	**42 cm^2**
14 cm	10 cm	8 cm	**96 cm^2**
20 m	12 m	5 m	**80 m^2**
13 cm	7 cm	10 cm	**100 cm^2**

Finding Height Given Area

Problem	Step 1	Step 2	Step 3	Answer
A=70 cm^2, a=12, b=8	sum=20	70 = ½ × 20 × h	70 = 10h	**h = 7 cm**
A=90 cm^2, a=15, b=5	sum=20	90 = ½ × 20 × h	90 = 10h	**h = 9 cm**

5 Finding Area of Circle

Formula: $A = \pi r^2$ (Use π = 22/7 or 3.142)

Problem	Step 1	Step 2	Step 3	Answer
r=14 cm	r^2 = 196	22/7 × 196	22 × 28	**616 cm^2**
r=21 cm	r^2 = 441	22/7 × 441	22 × 63	**1386 cm^2**
r=7 cm	r^2 = 49	22/7 × 49	22 × 7	**154 cm^2**
d=7 cm	r = 3.5	22/7 × 3.5^2	—	**38.5 cm^2**

Quick Application

Radius / Diameter	Area
r = 7 cm	**154 cm^2**
r = 14 cm	**616 cm^2**
r = 21 cm	**1386 cm^2**
d = 7 cm	**38.5 cm^2**
d = 14 cm	**154 cm^2**

Finding Radius Given Area

Problem	Steps	Answer
Area=154 cm^2	$22/7 \times r^2 = 154 \rightarrow r^2 = 49$	**r = 7 cm**
Area=616 cm^2	$22/7 \times r^2 = 616 \rightarrow r^2 = 196$	**r = 14 cm**

6 Real-World Application: Wall Area

Problem	Steps	Answer
Wall: length=8 m, height=3 m	Area = 8 × 3	**24 m^2**
Wall: length=12 m, height=4 m	Area = 12 × 4	**48 m^2**
Triangular wall: base=10 m, height=6 m	½ × 10 × 6	**30 m^2**

Example: Wall with a Door

Given	Calculation	Answer
Wall: 10 m × 4 m	Total = 10×4 = 40 m^2	
Door: 2 m × 1.5 m	Door = 2×1.5 = 3 m^2	
Area to paint	40 − 3	**37 m^2**

7 Real-World Application: Floor Tiling

Problem	Steps	Answer
Floor: 6 m × 5 m	Area = 30 m^2	
Tile: 30 cm × 30 cm	Area = 0.3×0.3 = 0.09 m^2	
Number of tiles	30 ÷ 0.09	**333.33 ≈ 334 tiles**

Quick Application

Floor Size	Tile Size	Number of Tiles
4 m × 3 m	20 cm × 20 cm	**300**
5 m × 4 m	25 cm × 25 cm	**320**
6 m × 4 m	30 cm × 30 cm	**267**

8 Real-World Application: Land Area

Problem	Steps	Answer
Rectangular plot: 20 m × 15 m	Area = 20×15	**300 m^2**
Circular garden: r = 7 m	Area = 22/7 × 49	**154 m^2**
Triangular plot: base=30 m, height=20 m	½ × 30 × 20	**300 m^2**

9 Compound Figures

Example: Square with Semicircle

Given	Steps	Answer
Square side = 14 cm	Square area = 14^2 = 196 cm^2	
Semicircle radius = 7 cm	Circle area = 22/7 × 49 = 154 cm^2	
Semicircle area	½ × 154 = 77 cm^2	
Total area	196 + 77	**273 cm^2**

Quick Application

Figure	Area
Square side 10 cm with semicircle radius 5 cm	**100 + 39.25 = 139.25 cm^2**
Rectangle 12 cm × 8 cm with triangle on top	**96 + 48 = 144 cm^2**

10 Application Checklist

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✅ APPLICATION CHECKLIST:

Step 1: Identify the shape
☐ Square, Rectangle, Triangle, Parallelogram, Trapezium, or Circle?

Step 2: Identify the measurements needed
☐ Side, length, breadth, base, height, radius, diameter?

Step 3: Choose the correct formula
☐ Square: a²
☐ Rectangle: l × b
☐ Triangle: ½ × base × height
☐ Parallelogram: base × height
☐ Trapezium: ½ × (sum of parallel sides) × height
☐ Circle: πr²

Step 4: Substitute the values
Step 5: Calculate the area
Step 6: Write the answer with correct units
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11 Self-Test — Application Questions

No.	Question	Answer
1	Side=8 cm. Find area of square.	**64 cm^2**
2	l=12 cm, b=5 cm. Find area of rectangle.	**60 cm^2**
3	base=10 cm, height=6 cm. Find area of triangle.	**30 cm^2**
4	base=8 cm, height=5 cm. Find area of parallelogram.	**40 cm^2**
5	a=10 cm, b=6 cm, h=4 cm. Find area of trapezium.	**32 cm^2**
6	r=7 cm. Find area of circle.	**154 cm^2**
7	d=14 cm. Find area of circle.	**154 cm^2**
8	Area=48 cm^2, base=8 cm. Find height of triangle.	**12 cm**
9	Area=60 cm^2, base=12 cm. Find height of parallelogram.	**5 cm**
10	Wall: 10 m × 3 m, door: 2 m × 1 m. Find area to paint.	**28 m^2**

Revision Tip: Practice applying area formulas to different shapes. Remember to use the correct units (cm^2, m^2) and always check your calculations!